Do Synthetic Personas Predict Real Audience Response? A Headline-Ranking Study with a Generic-Reader Baseline

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Abstract

Do demographic personas improve an LLM's prediction of audience engagement? We compare a fixed ten-persona panel with a generic-reader baseline on Upworthy headlines, with supplementary news and Reddit comparisons. The primary experiment collects Gemini 3.1 Flash-Lite and GPT-4.1 observations on the same 190-package exploratory cohort, selected with simultaneous winner screening. It allocates thirty scores per arm and variant within model, retains exact raw responses, and reports metric-specific valid and paired counts. Complete-package coverage is 188 for Gemini and 188 for GPT-4.1. The baseline-minus-panel Kendall τ_b gap is +0.222 (CI [+0.061, +0.385]) for Gemini and +0.257 (CI [+0.119, +0.391]) for GPT-4.1, on respectively 136 and 188 valid rank pairs. Expected top-1 is 39.1% versus 54.8% for Gemini and 42.0% versus 61.7% for GPT-4.1 (panel versus baseline). These intervals condition on the selected cohort and recorded generations and do not adjust for dependence between related experiments. The Upworthy archive documents a randomization-risk period. Excluding that period in a post-hoc sensitivity retains positive expected top-1 and lift gaps in both configurations, while Gemini tau-b and Spearman intervals include zero. Supplementary Reddit top-1 and captured lift favor the panel; these aggregate comparisons use unreconciled collection protocols and do not isolate dataset effects. A raw-replay artifact supports offline reproduction of the primary experiment. The results concern this panel, cohort, and protocol; they are not a held-out evaluation or evidence of universal persona harm. Provider request protocols differ across models, and no causal mechanism is isolated.

1 Introduction

LLM-based audience simulation offers a way to compare messages before live testing. Demographic conditioning can approximate aspects of survey samples [1], but a copy-ranking system must predict measured engagement. We study this narrower question with Upworthy headline experiments [16].

We compare a fixed ten-persona panel with a generic-reader baseline using a matched-budget experiment on a simultaneously screened Upworthy cohort. Both model configurations allocate the same number of scores to each arm and variant, preserve raw responses, and report distinct rank and decision metrics with explicit denominators. The primary analysis uses uniform expected credit for tied decisions and per-metric paired contrasts; a post-hoc period-exclusion sensitivity examines the archive's documented randomization risk. The cohort contains previously examined outcomes, so this is an exploratory evaluation rather than a held-out test.

Supplementary evidence includes aggregate-report comparisons and a GPT-4.1 scalar-checkpoint analysis. These use different collection budgets, winner screens, and tie rules and are reported separately from the primary comparison. Their evidential scope also differs: reexporting an aggregate does not reconstruct its raw observations, and replaying stored scalars cannot validate absent response texts. Raw-backed reproduction is available for the primary comparison; it does not validate the supplementary generations or the other datasets.

We ask (RQ1) how this panel and generic baseline compare within tested packages, (RQ2) how agreement varies with the observed outcome signal, and (RQ3) what the dataset, aggregation, seed, and model comparisons establish. This evaluates a specified harness rather than all persona methods or audience representativeness.

2 Related Work

Profile-conditioned models can approximate some aggregate survey distributions [1]; broader computational social science applications raise validity questions [25]. Interview-grounded agents reproduce General Social Survey responses at approximately 85% of participants' own test–retest consistency in the original version of Park et al. [19]. That result uses rich

individual grounding and a different outcome from clicks.

Other work documents option-order sensitivity [5], limited effects of persona attributes on subjective annotation, alongside modest significant gains in some settings [10], persona-related degradation on objective tasks [24], caricature [2], and biased outcomes from generated personas [15]. Reasoning biases in persona-assigned models are also documented [8]. These motivate a baseline, but do not explain this study's gap.

Upworthy [16] supports engagement analyses including negativity effects [20]. LOLA [23] studies LLM-assisted content experiments; differences in selection, models, and evaluation prevent a direct numerical comparison here. MIND [22] and Reddit reposts [14] are additional observational settings, not interchangeable headline experiments. Construct validity [4] distinguishes intent from behavior. Gamma [7] must also be distinguished from Kendall correlation and its treatment of ties [11, 12].

3 Sim-to-Real Validity Framework

A package contains variants with observed outcome rates and predicted scores (Figure 1). Upworthy was designed for randomized traffic, but the archive authors document randomization problems during part of 2013–2014 [17]. We report observed rate rankings; they are not automatically causal headline effects. Grouping distinct headlines can also combine image variants. MIND ranks different articles within a subcategory by clicks divided by logged article appearances, pooling user-level recommendation logs. Reddit groups reposts by image and subreddit, pools votes for identical titles, and uses upvotes divided by total upvotes plus downvotes. This is a vote-approval ratio, not CTR; vote totals are not exposure counts. Timing, recommendation, and exposure remain uncontrolled in these observational comparisons. Within-package ranking does not establish absolute CTR calibration or a causal copy effect in observational data.

Metric identity. For concordant and discordant pair counts C, D, Goodman–Kruskal gamma is

$$\gamma = (C - D)/(C + D),$$

discarding pairs tied in either variable [7]. It is undefined when $C + D = 0$ and differs from Kendall τ_b, which accounts for ties:

$$\tau_b = \frac{C - D}{\sqrt{(C + D + T_x)(C + D + T_y)}},$$

where T_x, T_y count pairs tied only in the respective variable [12]. Spearman uses average ranks [21]. Constant rank inputs yield undefined values.

The primary experiment uses uniform expected top-1 and bottom-1 credit: $|S_{\text{pred}} \cap S_{\text{real}}|/|S_{\text{pred}}|$ for the corresponding extreme sets. Captured lift averages observed rates across tied predicted maxima, then scales distance from the observed minimum by the observed range. Constant outcomes make lift undefined. Supplementary aggregate-report top-1 results use the first maximum and therefore depend on variant order.

4 Method Under Test

The study harness scores each headline with either a demographic persona or a generic reader. It operates independently of any commercial product. The fixed ten-persona JSON takes inspiration from Upworthy-era U.S. social-media demographics [9]. These descriptors do not specify a deterministic behavioral-profile generator or establish representativeness.

Elicitation and aggregation. The prompt requests JSON with a `click_intent` number in $[0, 1]$ (Appendix A). Scores are averaged across recorded observations:

$$\widehat{s}(v) = \frac{1}{|O_v|} \sum_{(i,d) \in O_v} y_{i,d}(v).$$

With complete draws this equals the mean over personas and draws. With missing draws it can differ from the supplementary aggregation mean, which weights draw-collapsed personas equally. The supplementary aggregation comparison evaluates these alternative rules.

Sampling and parsing. The primary experiment uses temperature 0.8, three draws per persona and thirty generic-reader draws per variant, with request seeds recorded for both providers. The supplementary Gemini comparison uses the same nominal draw allocation. The GPT-4.1 scalar-checkpoint analysis instead uses three draws per persona versus ten baseline draws; its requests sent no generation seed. Seeds alone do not guarantee identical provider responses over time.

Supplementary scalars were produced by a permissive parser that could extract a first number, clamp values, and accept booleans; event frequencies are unknown. The matched-budget collector accepts only a finite, non-boolean JSON number in range under the exact requested key. Invalid parses are recorded as terminal missing observations. Configuration hashes and

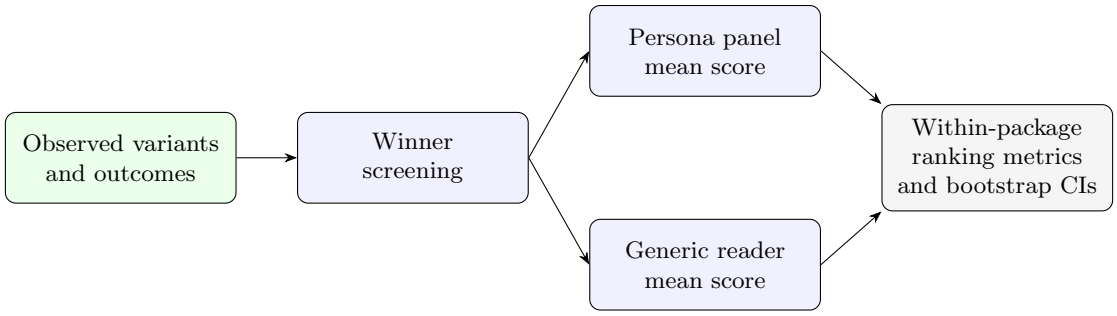


Figure 1: Comparison protocol. Outcomes determine selection; the two scoring arms are compared within selected packages. Supplementary aggregates use gamma and the top-versus-runner-up screen. The primary experiment and GPT checkpoint analysis additionally report tau-b and simultaneous selection. Selection does not establish that excluded packages are unpredictable.

separate protocol namespaces prevent mixing primary checkpoints with supplementary scores.

5 Experimental Setup

Corpus. Upworthy rows are grouped by experiment and distinct headline, aggregating identical text. Packages require at least two headlines and 3,000 impressions per headline; sampling uncertainty matters in online experiments [13]. The locked exploratory corpus has 6978 variants in 1695 packages. In these locked inputs, headline linebreaks were removed before grouping. One retained variant has such a linebreak in the source CSV; its package is outside both selected cohorts. Exact locked strings define the saved experiments, and the original manifests remain unchanged.

Archive allocation risk and sensitivity. The archive authors' 2024 report identifies a randomization-risk window from 25 June 2013 through 10 January 2014 inclusive [17]. The preserved CSV contains creation dates but neither the documented problem flag nor a risk indicator. Calendar-date overlap identifies 318 corpus packages, 83 supplementary-screen packages and 42 packages in the primary selected cohort. These counts mark window membership, not individually proven defects. Bonferroni screening cannot repair allocation problems.

A post-hoc offline sensitivity, specified after the primary collection, excludes an entire experiment if any row for that experiment in the preserved CSV has a creation date that falls in that inclusive window, using the original date string without inventing a timezone. Each model otherwise retains its complete primary-package policy and the same recorded generations, metrics and bootstrap procedure. Primary reports are preserved unchanged. Dates outside the window are not a certification of valid randomization; this sensitivity is not confirmatory or an independent replication.

Simultaneous winner screen. The primary experiment requires a unique observed maximum and compares it with every opponent using two-sided pooled z tests. For k variants the comparison family contains $m = k(k-1)/2$ unordered pairs. All winner–opponent comparisons must pass α/m, with $\alpha = 0.05$. The screening statistic is $\min(1, m \max_{j \neq w} p_{wj})$. Bonferroni's inequality motivates the adjustment [18]. The nominal bound is approximate and per package, conditional on independent binomial observations, fixed sampling horizons, and adequate normal approximation. It is not an across-package false-discovery guarantee or proof of a population winner. Repeated exposures and Reddit voting further limit these assumptions. In the locked corpus 190 packages pass, all within the supplementary selected set described below. A stricter screen does not demonstrate that removed packages had false winners. The archive documents editorial stopping based on accumulating results [16]. Fixed-horizon p-values need not remain calibrated under that stopping policy; Bonferroni does not repair invalid input p-values. The screen defines a reproducible cohort, without a verified five-percent winner-error guarantee for these archived tests.

Supplementary winner screen. The supplementary aggregate comparisons use the observed best variant and runner-up, comparing them using a one-sided pooled two-proportion z test at $p < 0.05$, selecting 399 packages. The direction was chosen from the data: even a two-arm null comparison can pass in either direction, so the nominal one-sided threshold does not

imply a five-percent false-positive rate. Comparing only the runner-up need not separate a winner from every opponent. Outcome selection changes the evaluation population. Packages failing a screen are not thereby unpredictable.

Matched-budget collection. We froze the protocol and its source before collection. Both Gemini 3.1 Flash-Lite and Azure GPT-4.1 score every variant of the same 190-package simultaneously screened exploratory cohort. This cohort was selected from previously examined outcomes and is not a held-out population. The matched-budget protocol is evaluated on this Upworthy cohort only; MIND, Reddit, other Upworthy splits, model tiers, seeds and aggregation rules are covered by the supplementary analyses, which do not establish prompt robustness.

Each variant is allocated three observations for each of ten fixed personas and thirty generic-reader observations. Package and variant order is shuffled with a recorded seed; baseline/persona requests are interleaved with randomized order within pairs. Temperature is 0.8, JSON output is requested, and output is limited to 64 tokens per choice. Gemini uses single-choice requests with MINIMAL thinking. Azure uses three indexed choices per request in both arms: ten requests per arm and variant. We match the number of retained scores within each model, not an identical transport or inference protocol across providers. Token use and monetary cost are not equalized. Choices in one request are not assumed independent. Request seeds, exact prompts, responses, returned model identities, usage, and failures are preserved. Provider aliases and best-effort seeds do not establish future replay of generations.

The primary analysis retains a package only when every planned observation is valid in both arms for all its variants. This also makes equal-persona and observation-weighted panel means identical. Failures are never replaced with additional semantic draws; bounded retries address transient transport failures only. Available-observation arm summaries are reported separately as sensitivity analyses, with their own coverage, in the `available_observation_sensitivity` field of the committed Gemini and Azure reports in `recollection/2026-10-31/`. The primary policy is stricter than the scalar-checkpoint analysis requirement of at least one valid score per variant. Operational compatibility probes are retained separately and excluded. An Azure prompt refusal initially triggered a global HTTP400 stop. Before comparative analysis, a documented technical amendment classified only the confirmed content-filter refusal type as terminal missing while allowing unrelated planned requests to continue. The blocked request is not retried or rewritten; filtering remains active. Original journal bytes, the pause, source versions and the amendment marker are preserved. Other permanent failures and budget stops remain blocking. This is a response to an observed technical incident, not a preregistered amendment. The fixed cohort closes before inspecting comparative performance; there is no outcome-dependent stopping or choice of model after observing results.

Estimation and denominators. Package metrics are averaged without size weighting. The displayed means describe the recorded packages. Bootstrap intervals summarize between-package resampling variability conditional on the selection rule and recorded scores; interpreting them for a population of similarly selected experiments additionally assumes that these packages are exchangeable independent units. Raw-backed reports include $n_{\text{total}}, n_{\text{valid}}, n_{\text{undefined}}$ and metric-specific package IDs. Each paired gap uses packages valid in both arms for that metric. Empty estimates are undefined, not zero. Percentile bootstrap CIs use 10,000 package resamples, seed 42 [6]. They condition on selection and recorded scores and omit generation, selection, and outcome-measurement uncertainty. These are pointwise estimation intervals, not simultaneous guarantees across metrics or models. Resampling treats experiment IDs as independent units. The archive also records follow-up tests with shared content [16]; related IDs may therefore be dependent. The intervals do not adjust for such dependence, and their coverage as population confidence intervals is not established. Supplementary aggregate coefficient-specific valid counts are not fully recorded; coverage totals are not those denominators.

Planning and robustness scope. We report coefficient estimates and intervals without assigning standardized effect-size categories [3]. The analysis does not establish prospective power for averages of within-package coefficients or equivalence. Time-stamped preregistration of all thresholds is not established.

Supplementary reports cover three Upworthy splits, MIND [22], Reddit [14], three Gemini models, seed runs, and aggregation. Prompt robustness is not evaluated by these artifacts. Reddit's manifest records many more selected packages than the evaluated subset; its subset-selection procedure is undocumented.

Table 1: Primary Gemini 3.1 Flash-Lite comparison on 188 complete packages from the 190-package selected cohort. Every retained variant has all planned observations in both arms. Counts are metric-specific valid panel/baseline/paired packages. Gap is baseline minus panel; decision gaps use percentage points. Percentile bootstrap CIs condition on this cohort and these recorded generations. Undefined coefficients are omitted with their denominators disclosed.

Metric	Panel [95% CI]	Baseline [95% CI]	Paired gap [95% CI]	n_{valid} p/b/pair
Goodman–Kruskal γ	0.082 [-0.031, 0.192]	0.381 [0.253, 0.501]	0.268 [0.101, 0.434]	185/137/136
Kendall τ_b	0.082 [-0.030, 0.191]	0.336 [0.214, 0.450]	0.222 [0.061, 0.385]	185/137/136
Spearman ρ	0.100 [-0.016, 0.214]	0.371 [0.248, 0.488]	0.236 [0.069, 0.405]	185/137/136
Expected top-1	39.1% [32.2, 46.3]	54.8% [49.4, 60.2]	15.7 pp [7.1, 24.0]	188/188/188
Expected bottom-1	43.4% [36.4, 50.3]	47.5% [41.4, 53.6]	4.2 pp [-4.1, 12.3]	188/188/188
Captured lift	0.458 [0.394, 0.522]	0.591 [0.541, 0.641]	0.133 [0.054, 0.210]	188/188/188

6 Results

Primary matched-budget comparison (RQ1). Tables 1 and 2 give the primary matched-budget experiment. Supplementary comparisons below retain their own protocols and evidence boundaries. The complete-package policy retains 188 Gemini packages and 188 GPT-4.1 packages; exclusions are respectively 2 and 2. Reports retain every planned observation and failure, with metric-specific valid and paired IDs.

For Gemini, panel and baseline Kendall τ_b are 0.082 and 0.336; the paired baseline-minus-panel gap is +0.222 (CI [+0.061, +0.385]) on 136 valid pairs. Expected top-1 is 39.1% versus 54.8%, with paired gap +15.7 percentage points (CI [+7.1, +24.0]).

For GPT-4.1, panel and baseline Kendall τ_b are 0.088 and 0.345; the paired gap is +0.257 (CI [+0.119, +0.391]) on 188 valid pairs. Expected top-1 is 42.0% versus 61.7%, with paired gap +19.7 percentage points (CI [+10.6, +28.7]). In Gemini, τ_b is undefined on 51 baseline packages and 3 panel packages. Marginal means therefore use different valid sets; their difference is not the paired gap. The paired rank comparison uses their valid intersection, while top-1 includes every complete package with expected credit for ties. Baseline-minus-panel CIs exclude zero for the rank metrics, top-1 and captured lift in both configurations; bottom-1 CIs include zero. These are pointwise conditional estimates. Gamma and τ_b have distinct rows and denominators; neither is substituted for the other. These conditional package-bootstrap intervals omit repeated-generation uncertainty and do not compare model families causally. The GPT scalar-checkpoint analysis uses a separate collection protocol.

Post-hoc randomization-window sensitivity. Table 3 excludes 40 Gemini and 41 GPT-4.1 complete primary packages in the documented risk window, leaving respectively 148 and 147 packages. Expected top-1 gaps remain +11.4 percentage points (CI [+1.7, +21.0]) and +22.4 points (CI [+12.2, +32.7]). Captured-lift gaps also remain positive with pointwise CIs excluding zero.

Gemini's Kendall τ_b gap becomes +0.157 (CI [-0.030, +0.346]) on 109 valid pairs; its tau-b and Spearman intervals include zero. GPT-4.1's tau-b gap is +0.272 (CI [+0.118, +0.425]) on 147 pairs. Thus the primary Gemini rank contrast is not robust across this cohort restriction and all rank estimators. Gamma remains a distinct conditional measure; it is not substituted for the uncertain tau-b result. Bottom-1 intervals include zero for both models. These are post-hoc, pointwise intervals, not an equivalence test.

Supplementary Upworthy comparison (RQ1). Table 4 and Figure 2 give supplementary aggregate estimates. The generic baseline has higher gamma (0.361 versus 0.084) and top-1 (49.2% versus 34.6%). These are conditional descriptions under the top-versus-runner-up screen and tie rule. Marginal CIs are not a paired test, and gamma is not assigned a calibrated medium-effect interpretation.

Figure 3 shows the supplementary screening statistic, not calibrated winner probabilities. The supplementary all-package panel gamma is 0.054 (CI [0.025, 0.083]), with the aggregate-report metric and parsing semantics.

Signal strength (RQ2). The pinned supplementary producer defines its separation index as $1 - p$, where p is the one-sided observed-best-versus-runner-up value, and correlates it with the panel's first-index top-1 hit. The stored Spearman association is 0.059 (CI [0.011, 0.107]) over 1678 packages, using 5,000 bootstrap resamples. Its code requires every variant to have a score and within-package Spearman to be defined, even for this top-1 analysis. The report alone

Table 2: Primary GPT-4.1 comparison on 188 complete packages from the 190-package selected cohort. Every retained variant has all planned observations in both arms. Counts are metric-specific valid panel/baseline/paired packages. Gap is baseline minus panel; decision gaps use percentage points. Percentile bootstrap CIs condition on this cohort and these recorded generations. Undefined coefficients are omitted with their denominators disclosed. Azure requested three indexed choices per call; choices are not assumed independent.

Metric	Panel [95% CI]	Baseline [95% CI]	Paired gap [95% CI]	n_{valid} p/b/pair
Goodman–Kruskal γ	0.088 [-0.026, 0.200]	0.345 [0.242, 0.445]	0.257 [0.119, 0.391]	188/188/188
Kendall τ_b	0.088 [-0.026, 0.200]	0.345 [0.242, 0.445]	0.257 [0.119, 0.391]	188/188/188
Spearman ρ	0.107 [-0.010, 0.222]	0.373 [0.265, 0.475]	0.266 [0.124, 0.404]	188/188/188
Expected top-1	42.0% [35.1, 48.9]	61.7% [54.8, 68.6]	19.7 pp [10.6, 28.7]	188/188/188
Expected bottom-1	43.1% [36.2, 50.0]	52.1% [44.7, 59.0]	9.0 pp [0.0, 17.6]	188/188/188
Captured lift	0.467 [0.400, 0.534]	0.654 [0.590, 0.719]	0.187 [0.103, 0.271]	188/188/188

Table 3: Post-hoc sensitivity excluding the archive's documented randomization-risk window (25 June 2013–10 January 2014 inclusive) from each model's complete primary cohort. Primary reports and recorded generations remain unchanged. All six metrics are shown; gaps are baseline minus panel, decision gaps are percentage points, and counts are metric-specific valid pairs. Pointwise percentile bootstrap CIs condition on retained packages and generations; remaining tests are not certified defect-free.

Model (complete packages)	Metric	Paired gap [95% CI]	n_{paired}
Gemini (148)	Goodman–Kruskal γ	0.201 [0.010, 0.396]	109
Gemini (148)	Kendall τ_b	0.157 [-0.030, 0.346]	109
Gemini (148)	Spearman ρ	0.170 [-0.022, 0.367]	109
Gemini (148)	Expected top-1	11.4 pp [1.7, 21.0]	148
Gemini (148)	Expected bottom-1	3.1 pp [-6.4, 12.6]	148
Gemini (148)	Captured lift	0.091 [0.001, 0.181]	148
GPT-4.1 (147)	Goodman–Kruskal γ	0.272 [0.118, 0.425]	147
GPT-4.1 (147)	Kendall τ_b	0.272 [0.118, 0.425]	147
GPT-4.1 (147)	Spearman ρ	0.292 [0.133, 0.450]	147
GPT-4.1 (147)	Expected top-1	22.4 pp [12.2, 32.7]	147
GPT-4.1 (147)	Expected bottom-1	8.8 pp [-0.7, 18.4]	147
GPT-4.1 (147)	Captured lift	0.213 [0.117, 0.307]	147

cannot identify which excluded packages lacked scores or had undefined ranks. This index mixes outcome separation with exposure, and top-1 difficulty also varies with the number of variants; the association is unadjusted for those factors. It is not evidence of an intrinsic predictability effect. Missing supplementary predictions prevent recomputing the analysis under simultaneous screening and expected tie credit; the primary selected-cohort experiment does not estimate this all-package association.

Dataset comparisons (RQ3). Table 5 shows higher baseline gamma on the three Upworthy splits and a smaller difference on MIND, where packages compare articles. On Reddit, the panel has higher top-1 (40.3% versus 30.2%) and lift (0.452 versus 0.333). The available summaries show different directions, but unreconciled collection and subset differences prevent attributing this variation to the dataset alone. Reddit gamma CIs include zero; this establishes neither equivalence nor absence of a title-related signal. Posting time and visibility were not tested as explanations. Subset-selection and coverage limitations further restrict interpretation. The aggregate summary does not record draw counts or evaluated package IDs, so cross-dataset protocol equivalence is not independently verified by those aggregates.

Seed, model, ties, and aggregation. The supplementary three-seed SDs (0.025 for gamma; 0.008 for top-1) use the population convention over those seeds, not sample SDs or precise general-variability estimates. The baseline has higher gamma on three tested Gemini tiers (Table 9, Appendix A); evaluated

Table 4: Supplementary Upworthy exploratory estimates under the top-versus-runner-up screen. Panel $n_{total} = 399$; baseline $n_{total} = 398$; correlation-specific valid counts are unavailable. The stored rank coefficient is Goodman–Kruskal gamma, not Kendall tau. Top-1 uses first-index tie breaking. Random top-1 reference 30.4%. Marginal 95% bootstrap CIs; the reported screen and decision tie rule are retained.

Method	Top-1 [95% CI]	Goodman–Kruskal γ [95% CI]	Spearman ρ [95% CI]
No-persona baseline	49.2% [44.2, 54.3]	0.361 [0.289, 0.430]	0.350 [0.281, 0.417]
Persona panel	34.6% [29.8, 39.1]	0.084 [0.020, 0.147]	0.104 [0.035, 0.172]

Table 5: Supplementary estimates under the top-versus-runner-up screen and first-index tie breaking. n_{total} is panel/baseline coverage, not the correlation-specific valid count (unavailable). Lift is the fraction of the observed outcome-rate range captured by the selected variant (panel/baseline). The panel exceeds the baseline on Reddit lift and top-1 (discussed in text); near-zero gamma does not establish equivalence or unpredictability. Gap is the difference of reported marginal gamma means, not a paired contrast. The summary does not retain draw counts or evaluated package IDs; the exploratory row differs from Table 4, and protocol equivalence cannot be verified from the aggregates alone.

Dataset	n_{total} p/b	Persona γ	Baseline γ	Gap	Lift p/b
Upworthy exploratory (2013–15)	399/398	0.079	0.361	+0.282	0.45/0.58
Upworthy holdout (2013–15)	402/401	0.100	0.398	+0.298	0.48/0.56
Upworthy confirmatory (2013–15)	1866/1864	0.070	0.393	+0.323	0.44/0.55
MIND news (2019)	74/74	0.177	0.214	+0.038	0.47/0.51
Reddit titles (2008–12)	600/599	-0.009	-0.026	−0.017	0.45/0.33

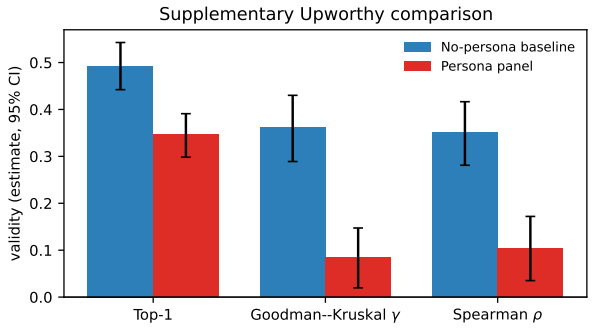


Figure 2: Supplementary comparison under the top-versus-runner-up screen. Coefficient-specific valid counts are unavailable. The displayed rank coefficient is gamma.

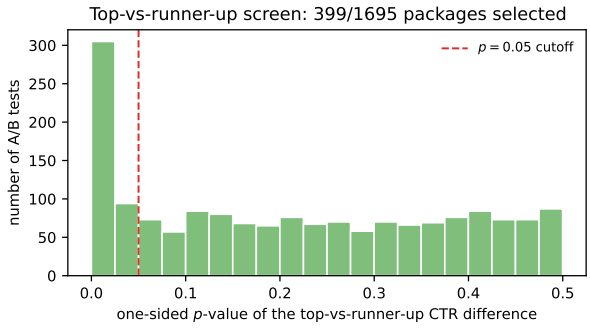


Figure 3: Supplementary one-sided top-versus-runner-up screen selecting 399 packages. Selection of direction from observed rates invalidates a literal one-sided false-positive-rate interpretation. Nonsignificant packages need not be unpredictable.

IDs are not retained to verify a shared subset. This cannot establish universal model or prompt robustness.

The tie report rounds scores before counting constant packages; the rank code uses exact equality. That count cannot recover excluded gamma values. Undefined coefficients cannot be silently assigned zero, and the reports do not justify a conservative-lower-bound claim for the gap.

Table 6 preserves six supplementary aggregation rules, all with lower reported gamma than the baseline. Equal-persona and observation-weighted means can differ with missing draws. The reports do not establish why their top-1 results differ, or whether bias, sampling noise, scale, or another factor explains relative performance.

GPT-4.1 scalar-checkpoint analysis. The recovered checkpoint contains scalars, not original responses. Offline replay of the pinned producer reproduces all published cross-family report fields within

Table 6: Supplementary aggregation comparison (n_{total} = 399 for panel rules). The baseline has 398 covered packages; correlation-specific valid counts are unavailable. The mean row weights draw-collapsed personas equally, whereas the primary collector weights recorded observations equally. These estimands can differ when draws are missing; the stored reports do not identify the cause of their top-1 discrepancy. No tested rule reaches the baseline's reported gamma. This descriptive comparison does not identify a failure mechanism. Top-versus-runner-up screen, first-index tie breaking, 95% marginal bootstrap CIs.

Aggregation	Goodman–Kruskal γ [95% CI]	Top-1 [95% CI]
No-persona baseline	0.361 [0.289, 0.430]	49.2% [44.2, 54.3]
Equal-persona mean	0.084 [0.020, 0.147]	34.8% [30.1, 39.3]
Panel median	0.141 [0.076, 0.205]	39.3% [34.6, 44.1]
Trimmed mean (20%)	0.081 [0.017, 0.144]	34.3% [29.6, 38.8]
Per-persona z-mean	0.064 [0.001, 0.128]	33.6% [29.1, 38.3]
Borda (rank fusion)	0.081 [0.018, 0.146]	35.3% [30.8, 40.1]
Condorcet (pairwise)	0.079 [0.014, 0.146]	35.8% [31.3, 40.6]

numerical tolerance. Collection coverage is 399 panel and 395 baseline/paired packages. Thirty panel observations versus ten baseline draws and no generation seed make it an unequal-budget comparison.

Table 7 applies simultaneous screening and tie-aware estimators to those scores. Coverage becomes 190 panel, 189 baseline, and 189 paired packages. Baseline Kendall τ_b is 0.325 versus 0.049. Paired CIs for the displayed gaps exclude zero, supporting a positive baseline gap in this checkpoint analysis. It is not an independent replication or a budget-matched estimate, and does not validate raw parsing. Gamma and tau-b remain separate. The supplementary Gemini observations are unavailable for applying these estimators; the primary Gemini comparison uses its own retained generations and configuration.

7 Discussion and Limitations

The comparisons make a generic-reader baseline necessary for this task. A persona panel adds assumptions about audience composition and behavior that require predictive validation. The primary matched-score comparisons favor the baseline in valid rank comparisons, expected top-1 and captured lift, but do not establish a bottom-1 advantage. Gemini rank means exclude constant outputs with explicit arm-specific counts; top-1 retains those packages through expected tie credit. Supplementary Upworthy reports and the GPT checkpoint analysis also favor the baseline; supplementary Reddit decision metrics favor the panel. Excluding the documented randomization-risk period preserves the primary top-1 and lift contrasts but leaves Gemini tau-b and Spearman uncertain. The ranking conclusion therefore depends on cohort and estimator, not just model configuration.

Mechanism remains open. The baseline may use general headline knowledge, while role-play may add unhelpful conditioning. Neither explanation was isolated experimentally. Similar score spreads, selected ranking examples, and rank-only aggregation failures cannot distinguish bias from noise or identify a mechanism. The balanced, interleaved primary collection tests the panel/baseline comparison with retained raw responses, but does not separately manipulate demographic content, role-play, or response noise. It therefore does not by itself identify those mechanisms. The archive measures clicks on Upworthy's website, while the prompt frames a social-media reader in 2014 and omits the image. This platform/time framing is not an exact reconstruction of the test audience. Intent versus engagement remains a construct-validity limitation [4].

Ground truth and generalization. The Upworthy archive documents a period of potentially invalid randomization [17]; our selected cohort includes affected-window tests. Our predictions rank recorded outcome rates, and neither an adjusted screen nor model responses establishes causal headline effects. The post-hoc exclusion sensitivity addresses one known risk window, not every possible allocation or image confound. Editorial stopping and shared content across follow-up experiments [16] add distinct risks: the screen does not correct sequential monitoring, and the package bootstrap does not cluster related experiment IDs. The preserved aggregate outcomes do not reconstruct the stopping trajectories, and dependence has not been quantified by an article-level cluster analysis. Nominal intervals may understate uncertainty. Simultaneous screening does not make outcome measurements certain. Exclusion changes the target population and relative task difficulty. Failure to reject equality does not imply no predictor can recover an ordering. MIND and Reddit are observational; correlated exposures, time, image, audience mix, and topic may matter. Training contamination and generalization to current audiences or other formats are not excluded by the archive corpus.

Reproduction limits. Aggregate reports allow re-export, not complete verification of their generating process. Supplementary Gemini observations and some aggregate-report producers were not recovered.

Table 7: GPT-4.1 scalar-checkpoint analysis on recorded post-parser values. Simultaneous winner screen and uniform expected tie credit; marginal percentile bootstrap CIs. Gap is baseline minus panel on paired packages; top-1 gap is in percentage points. Counts are per-metric valid panel/baseline/paired. Original responses are unavailable, so this does not validate the collection parser. Sampling remains 30 panel observations versus 10 baseline draws per variant.

Metric	Panel [95% CI]	Baseline [95% CI]	Paired gap [95% CI]	n_{valid} p/b/pair
Goodman–Kruskal γ	0.049 [-0.061, 0.159]	0.325 [0.223, 0.429]	0.270 [0.132, 0.407]	190/189/189
Kendall τ_b	0.049 [-0.061, 0.159]	0.325 [0.222, 0.429]	0.270 [0.132, 0.407]	190/189/189
Expected top-1	39.5% [32.6, 46.3]	60.6% [53.7, 67.7]	20.9 pp [11.1, 30.4]	190/189/189

The GPT checkpoint supports replay and partial reanalysis, but range-checking stored scalars cannot retroactively validate the permissive parser. The primary strict-parser observations have complete raw lineage and matched within-model score budgets; they are not mixed with supplementary scores. Azure multi-choice and Gemini single-choice requests differ operationally. Returned aliases are not immutable model snapshots, and request seeds do not guarantee determinism. A single collection session does not estimate provider drift or generation variability. Content filtering can cause exclusions associated with content or arm; the Azure technical pause also divides its collection in time. The outcome-selected cohort limits its generalization. Prompt robustness, deterministic demographic-to-panel construction, full Reddit subset lineage, and preregistration chronology remain evidence gaps.

The artifact package distinguishes aggregate reexport, scalar-checkpoint analysis and raw-response replay, so each reported comparison can be assessed within its available evidence.

8 Conclusion

This fixed demographic persona panel should be compared with a generic-reader LLM baseline. The primary raw-retained comparison favors the baseline on valid rank comparisons, expected top-1 and captured lift in both tested configurations. In the post-hoc sensitivity excluding the archive's documented randomization-risk period, top-1 and lift gaps retain positive pointwise intervals, but Gemini tau-b and Spearman intervals include zero. The rank advantage is therefore not uniformly supported across these cohorts and estimators. In the primary comparison, expected top-1 baseline-minus-panel gaps are +15.7 percentage points for Gemini and +19.7 for GPT-4.1. Bottom-1 intervals include zero; a blanket advantage across metrics is not established. Supplementary Reddit top-1 and lift favor the panel.

The comparison uses distinct rank metrics, expected tie credit, and explicit valid and paired counts. We provide a simultaneous winner screen and an offline raw-replay artifact in the accompanying local replication bundle, prepared for deposition. Rank estimates condition on valid coefficients, and all intervals condition on selected packages and recorded generations. Related-test dependence and editorial stopping remain unadjusted, limiting interval calibration and winner-screen interpretation. Supplementary Gemini observations remain unavailable. This study neither establishes a general causal mechanism nor shows universal persona harm.

Data and Code Availability

The package at `https://github.com/alemaiorano/sim-to-real-validity` includes sources, exporters, supplementary aggregate reports, the fixed panel, and original corpus manifests. Reexporting these reports reproduces their displayed numbers, not their generation from raw observations. Original reports are hash-pinned and unchanged. GPT-4.1 checkpoint analyses, input hashes, and offline replay verification are in `reanalysis/2026-10-31/`. A separate replication bundle retains the GPT scalar checkpoint, exact exploratory inputs, and the primary matched-budget experiment: all raw response bytes, request and observation plans, immutable journal snapshots, coverage, package metrics, and source history. Committed primary aggregate reports and hash provenance are in `recollection/2026-10-31/`. Offline raw replay needs no provider account. Supplementary Gemini observations remain missing; primary generations do not restore them. `REPLICATION.md` distinguishes these evidence tiers. The accompanying replication bundle is prepared locally for deposition; it is not yet available at the public repository URL.

Ethics and Conflict of Interest

The study uses public archive data and synthetic personas; no new human subjects were recruited. The author has a commercial interest in copy-simulation tools. This evaluates a study harness rather than a specific product. Disclosure, adverse results, explicit limitations, and available artifacts and the accompanying local bundle support scrutiny; an unverified preregistration chronology is not presented as a safeguard.

AI Tools Disclosure

This research used AI-assisted development tools for manuscript preparation and code development under human oversight and accountability.

- **Language models:** Claude (Anthropic, via Claude Code) and GPT-family models (OpenAI, via Codex) supported code implementation, analysis, literature checks, and manuscript preparation and review. Automated manuscript feedback also supported the review. The Google Gemini and Azure GPT-4.1 configurations evaluated in the study are experimental components, as described in the methods and supplementary analyses; their outputs are study observations.
- **Web search:** MCP Tavily and web search tools supported literature review and fact-checking during manuscript preparation.

The author retains full responsibility for the scientific arguments, methodology, analysis, citations, and conclusions. Numerical and literature claims are checked against retained artifacts and primary sources; AI-generated suggestions and automated reviews do not constitute independent empirical validation.

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A Persona Panel, Prompt, and Reproducibility Details

Fixed panel. The ten-persona JSON is the reproducible input. Demographic inspiration [9] is not a deterministic recipe for recreating behavioral descriptions. The unweighted panel is not an estimated population sample.

First-person template.

> You are simulating a specific person reacting to a news headline on social media in 2014.
> Person: `{description}` (age `{age_group}`, `{gender}`, mainly on `{platform}`, news engagement: `{news_engagement}`.)
> Headline: `"{headline}"`
> As this exact person, how likely are you to click this headline? Respond ONLY with JSON: `{"click_intent": <number between 0 and 1>}`.

The primary panel condition uses this first-person template; the generic-reader condition uses the baseline template below. Each is sent as a single user

message, without a system message. The exact templates, including spacing, quotation marks and implemented alternative framings, are authoritative in the versioned collector. A prompt implementation alone does not establish a completed experiment. The primary collection uses fixed 2014 framing for all headlines, including those created outside that year; it is not a date-specific audience reconstruction.

Generic-reader baseline template.

> You are a typical U.S. social-media news reader in 2014.
> Headline: `"{headline}"`
> How likely are you to click this headline? Respond ONLY with JSON: `{"click_intent": <number between 0 and 1>}`.

Both templates retain the U.S./2014 framing. The baseline specifies a generic reader role; its distinction from the panel is the absence of the fixed demographic profiles, not the absence of all role conditioning. Literal spacing, persona fields, and request prompts are retained in the collector and raw request plans. The interpolated persona fields are description, age group, gender, platform, and news engagement, all shown in Table 11. Political lean is stored metadata and is not separately interpolated.

Coverage and checkpoints. Coverage policies differ by analysis. The GPT scalar-checkpoint analysis requires at least one recorded score for every variant, with means over available observations. The primary matched-budget experiment requires all planned scores in both arms for a retained package. Terminal failures, request/choice indices, and exclusions are preserved. Configuration and source hashes prevent mixing protocols. Each Azure request has one seed shared by its three returned choices; there is no invented per-choice seed. GPT scalar-checkpoint collection requests did not send a seed.

Raw artifact and replay. The primary collector appends and fsyncs a request record before dispatch and retains the exact response bytes, status, usage, version, and parsing outcome afterward. Hash-chained journals are synchronously mirrored. Unknown outcomes after interruption remain unknown and are not silently regenerated. Offline analysis validates the request plan, input hashes, response hashes, and journal chain, then reparses the same immutable snapshot whose hash identifies the report. The replication bundle contains that snapshot, all inputs, request plans, aggregate reports, source history, and the retained GPT scalar checkpoint. Replaying saved bytes requires no provider account and does not assert that future generations will match.

Supplementary outcome construction. MIND aggregates the supplied news and behavior logs by article ID, counting appearances in recommendation lists and click labels across users. Articles need at least 200 appearances; within each subcategory, at most ten are retained by appearance count, with at least two per package. Reddit aggregates identical titles within image/subreddit pairs, requires at least 300 total votes per title, and retains at most ten titles by vote count. Upvotes are the numerator and total upvotes plus downvotes the denominator, regardless of the schema's `clicks/impressions` names. These inputs cover the supplied logs, not a matched exposure window or common audience. Original manifests and the pinned corpus builders specify the transformations; missing evaluated IDs prevent fully reconciling the supplementary samples.

Supplementary pilot and model comparison. Tables 8 and 9 retain the pilot and cross-tier gap; shared package IDs are not retained in those summaries. Overlap of pilot CIs does not establish equivalence. The supplementary Upworthy comparison uses Gemini 3.1 Flash-Lite. No cost comparison is available from the pilot report.

Headline-ranking examples and panel. Table 10 shows supplementary reported cases where the baseline found the observed winner and the panel did not; they are not representative causal evidence. Table 11 presents the fixed panel descriptions. Full fields and grounding notes remain in `data/persona_panel.json`.

Table 8: Supplementary pilot with reported coverage $n_{total} = 100$ packages per model. Evaluated IDs and coefficient-specific valid counts are unavailable; a shared subset cannot be verified. The intervals overlap; this is not an equivalence test. The stored pilot contains no cost measurements and does not establish cost-effectiveness. 95% bootstrap CIs.

Model	Top-1	Goodman–Kruskal γ	Spearman ρ
`gemini-2.5-flash`	0.390 [0.29, 0.49]	0.116 [-0.01, 0.24]	0.133 [-0.01, 0.27]
`gemini-3.1-flash-lite`	0.330 [0.24, 0.42]	0.035 [-0.10, 0.17]	0.051 [-0.09, 0.19]
`gemini-3.5-flash`	0.390 [0.29, 0.49]	0.122 [-0.01, 0.25]	0.139 [-0.00, 0.28]

Table 9: Supplementary cross-model comparison with reported coverage ($n_{total} = 100$ per condition). The baseline has higher reported gamma on these three tested Gemini models. This does not establish a model-independent effect. Evaluated IDs and correlation-specific valid counts are unavailable; a common subset cannot be verified from this summary. Gap is the difference of reported marginal gamma means, not a paired contrast.

Model	Persona γ	Baseline γ	Gap
`gemini-3.1-flash-lite`	0.035	0.401	+0.367
`gemini-2.5-flash`	0.116	0.219	+0.103
`gemini-3.5-flash`	0.122	0.404	+0.282

Table 10: Supplementary reported examples of observed winning headlines that the no-persona baseline ranked first but the persona panel did not (of 126 such cases). These are selected illustrations, not evidence of a systematic causal mechanism.

Real winning headline	CTR	#var
Bookmark This For When You're Sad And Instantly Be 100 Times Happier	1.38%	4
10 Amazing International Photos That Will Give You Some Perspective	1.92%	4
If You Saw Someone Bullying A Kid For Being Gay Would You Speak Out...	1.11%	2
If You Watch Hilary Swank Annoy A Whiny Senator, $1 Will Go To Pun...	1.67%	3
Seven Wonderful People Share Secrets That Are Killing Them. This C...	1.99%	4
A 9-Year-Old Asks McDonald's CEO A Simple Question. She's Not Lovin...	3.23%	4
Did These Words Really Come Out Of Conan's Mouth?	4.45%	3
We Use This Word So Often We Probably Don't Realize We're Saying It...	1.68%	4
Did You Know You Can Go For A Swim In Google Earth?	0.88%	3
Cookie Monster Is In Jail? I'm Actually Really Happy About The Reas...	2.30%	4

Table 11: Prompt fields of the fixed persona panel (P=10), inspired by Upworthy-era audience U.S. social-media demographics [9], with the verbatim behavioural descriptions used in the prompt. F/M denotes female/male. Political lean is retained as metadata in the replication package (`data/persona_panel.json`) and is not separately interpolated.

ID	Demographics	News engagement	Behavioural description (verbatim)
p01	18-29, F, Facebook	incidental	29-year-old woman, scrolls Facebook on her phone during breaks; clicks human-interest and feel-good stories shared by friends; rarely seeks news directly.
p02	18-29, M, Twitter	active	24-year-old man, heavy Twitter user, follows journalists and culture accounts; clicks surprising or contrarian takes; low tolerance for clickbait he's seen before.
p03	30-49, F, Facebook	incidental	38-year-old working mother, checks Facebook in the evening; drawn to headlines about family, health, and local impact; shares uplifting or outrage-inducing stories.
p04	30-49, M, Reddit	active	41-year-old man, browses Reddit and Facebook; skeptical of sensational headlines, clicks when a headline promises concrete information or a strong narrative payoff.
p05	50-64, F, Facebook	incidental	57-year-old woman, mostly on Facebook; clicks emotionally resonant stories and anything about social causes she follows; less drawn to internet-slang headlines.
p06	50-64, M, Facebook	moderate	60-year-old man, reads news on Facebook and cable; clicks political and economic headlines, distrusts overtly hyped 'you won't believe' framing.
p07	18-29, F, YouTube	incidental	22-year-old college student, video-first, on YouTube and Instagram; clicks visually evocative or identity-relevant headlines; very sensitive to authenticity.
p08	30-49, M, Facebook	incidental	45-year-old man, casual Facebook user; clicks curiosity-gap headlines that promise a quick interesting fact, especially maps, lists, and 'what happened next' stories.
p09	65+, F, Facebook	moderate	68-year-old retiree, active on Facebook with family; clicks heartwarming, nostalgic, or cause-driven stories; avoids headlines that feel like tricks.
p10	30-49, F, Twitter	active	34-year-old professional, follows news and advocacy on Twitter; clicks data-driven or justice-oriented angles; fatigued by repetitive viral formats.